STANDATA interpretation 23-BCI-008

Building

Smoke detectors in smoke or fire/smoke damper installation

Date Issued: October 2024

Purpose

Clarifies whether the National Building Code – 2023 Alberta Edition (NBC(AE)) requires a smoke detector as part of a smoke or fire/smoke damper operation to be connected to the fire alarm system (FAS).

Discussion

In accordance with Sentence 3.1.8.11.(3) smoke damper and combination smoke/fire dampers must be configured to close automatically upon a signal from an adjacent smoke detector.

The requirements specified in Sentence 3.1.8.11.(3) pertain exclusively to the location of the smoke detector. This provision does not require adhering to any additional fire alarm system requirements. The smoke detector mentioned here is meant to operate as part of the damper system and is not intended to connect to or require the need for a FAS.

As per NBC(AE) Sentence 3.1.8.5.(3) smoke dampers must conform to NFPA 105, Article 6.3.2.4.

Where smoke detectors are used to control smoke dampers or fire/smoke dampers the detectors are required to comply with NFPA 72.

The confusion may stem from the definition of a smoke detector in the NBC(AE) where it is defined as a type of fire detector. A fire detector is defined as a device that sends a signal, to the fire alarm system.

Although the NBC(AE) provides a definition for a fire detector, there are no mandated requirements for a smoke detector serving a smoke damper or combination smoke/fire damper to send a signal to the FAS.

Code References

Division A, Sentence 1.4.1.2.(1)

> ***Smoke detector*** means a *fire detector* designed to operate when the concentration of airborne combustion products exceeds a predetermined level.
>
> ***Fire detector*** means a device that detects a fire condition and automatically initiates an electrical signal to actuate an *alert signal* or *alarm signal* and includes *heat detectors* and *smoke detectors*.

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Article 3.1.8.4

3.1.8.4. Determination of Ratings and Classifications

1) Except as permitted by Sentences (2) and 3.1.8.16.(1), the *fire-protection rating* of a *closure* shall be determined in accordance with
   a) CAN/ULC-S104, "Standard Method for Fire Tests of Door Assemblies,"
   b) CAN/ULC-S106, "Standard Method for Fire Tests of Window and Glass Block Assemblies," or
   c) CAN/ULC-S112, "Standard Method of Fire Test of Fire Damper Assemblies."
(See Articles 3.1.8.17. to 3.1.8.19. for additional requirements for *closures*.)

2) Except as permitted by Sentence 3.1.8.12.(1), the *fire-protection rating* of a *closure* shall conform to Table 3.1.8.4. for the required *fire-resistance rating* of the *fire separation*.

3) The leakage rate of smoke dampers and combination smoke/*fire dampers* shall
   a) be determined in accordance with the applicable provisions in CAN/ULC-S112.1, "Standard for Leakage Rated Dampers for Use in Smoke Control Systems," and
   b) conform to Class I, II or III of that standard.

Sentence 3.1.8.5.(3)

3.1.8.5. Installation of Closures

3) Except as otherwise specified in this Part, every smoke damper or combination smoke/*fire damper* used as a *closure* in a required *fire separation* shall be installed in conformance with NFPA 105, "Standard for Smoke Door Assemblies and Other Opening Protectives."

Sentence 3.1.8.7.(2)

3.1.8.7. Location of Fire Dampers and Smoke Dampers

2) Except as provided in Article 3.1.8.9., a smoke damper or a combination smoke/*fire damper* shall be installed in conformance with Article 3.1.8.11. in ducts or air-transfer openings that penetrate an assembly required to be a *fire separation*, where the *fire separation*
   a) separates a *public corridor*,
   b) contains an egress door referred to in Sentence 3.4.2.4.(2),
   c) serves an *assembly*, *care*, *treatment*, *detention* or *residential occupancy*, or
   d) is installed to meet the requirements of Clause 3.3.1.7.(1)(b) or Sentence 3.3.3.5.(4).

Objective
OS1
Intent 1. To limit the probability that smoke will spread from one fire compartment to another fire compartment through openings in a fire separation, which could lead to harm to persons in the other fire compartment.

Objective
OP1
Intent 1. To limit the probability that smoke will spread from one fire compartment to another fire compartment through openings in a fire separation, which could lead to damage to the portion of the building in the other fire compartment.

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Sentence 3.1.8.11.(3)

3.1.8.11. Installation of Smoke Dampers

3) Except as required by a smoke control system, smoke dampers and combination smoke/*fire dampers* shall be configured so as to close automatically upon a signal from an adjacent *smoke detector* located as described in CAN/ULC-S524, "Standard for Installation of Fire Alarm Systems," within 1.5 m horizontally of the duct or air-transfer opening in the *fire separation*

a) on both sides of the air-transfer opening, or

b) in the duct downstream of the smoke damper or combination smoke/*fire damper*.

NFPA-105-2013, Chapter 6 Installation, Testing, and Maintenance of Smoke Dampers

6.3.2.4
Installation of combination fire and smoke dampers shall be in accordance with the installation of fire dampers in NFPA 80, Section 19.2.

6.3.2.5
Smoke detectors used to control smoke dampers or combination fire and smoke dampers shall be spaced and installed per the requirements of *NFPA 72, National Fire Alarm and Signaling Code.*

NFPA-72-2019, 21.7 Heating, Ventilating and Air-Conditioning (HVAC) Systems

21.7.2*
If connected to the fire alarm system serving the protected premises, all detection devices used to cause the operation of HVAC systems, smoke dampers, fire dampers, fan control, smoke doors, or fire doors shall be monitored for integrity in accordance with Section **12.6** and **23.8.5.3**.

Where the devices are connected to the fire alarm or signaling system, the wiring to these devices is required to be monitored for integrity in the same way as any other system wiring. Stand-alone detectors, including 120 VAC-powered, single-station smoke detectors used to control HVAC equipment, that are not connected to the fire alarm or signaling system cannot be monitored for integrity. For the 2022 edition of the Code, a reference was added to **23.8.5.3**, which provides requirements for initiating devices with separate power and signaling wiring.

FAQ Does NFPA 72 require detection devices used for HVAC system control to be connected to the building fire alarm or signaling system?

NFPA 72 does not specifically require detection devices used for HVAC system control to be connected to the building fire alarm or signaling system.

21.7.4.1
Smoke detectors mounted in the air ducts of HVAC systems in a fire alarm system without a constantly attended location or supervising station shall be permitted to initiate an alarm signal.

21.7.4.2
Smoke detectors mounted in the air ducts of HVAC systems shall be permitted to initiate an alarm signal where required by other governing laws, codes, or standards.

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Application

This interpretation applies to smoke detectors used for initiating the operation of smoke dampers or fire/smoke dampers in HVAC systems for buildings constructed in conformance with the NBC(AE).

Interpretation

1. Smoke dampers or fire/smoke dampers are intended to control the spread of smoke or fire/smoke in specified locations.

2. Smoke detectors used to initiate the operation of smoke dampers or fire/smoke dampers are permitted to be either supervisory or alarm initiating devices.

3. Smoke detectors located as described in CAN/ULC-S524 and only intended to activate smoke dampers and combination smoke/fire dampers are not required to be connected to a fire alarm system if the system is not required in the building by Article 3.2.4.1. Sentence 3.1.8.11.(3) does not extend the requirement for a fire alarm system beyond the requirement in Article 3.2.4.1.

4. Smoke detectors used as an initiating device for smoke or fire/smoke dampers are not required to be connected to a building FAS.

This INTERPRETATION is applicable throughout the Province of Alberta.

Unless stated otherwise, all Code references in this STANDATA are to Division B of the National Building Code – 2023 Alberta Edition

Issued by the Provincial Building Administrator

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